

Buffers

Answer key with worked solutions

A. Understanding

1. What a buffer is.

A solution that resists changes in pH when small amounts of acid or base are added. It contains a **weak acid and its conjugate base** (or a weak base and its conjugate acid).

2. Which are buffers?

- (a) **No:** HCl is a strong acid and Cl^- is too weak a base to remove added acid.
(b) **Yes:** weak acid + its conjugate base.
(c) **Yes:** weak base + its conjugate acid.
(d) **No:** NaOH is a strong base with no weak acid partner.

3. How the buffer works.

- (a) $\text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+ \longrightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$
(b) $\text{CH}_3\text{COOH} + \text{OH}^- \longrightarrow \text{CH}_3\text{COO}^- + \text{H}_2\text{O}$

B. Buffer pH

4. 0.10 M acid, 0.15 M acetate.

$$\text{pH} = 4.74 + \log(0.15 \text{ M} / 0.10 \text{ M}) = 4.74 + 0.18 = \mathbf{4.92} \text{ (the units of M cancel inside the log)}$$

5. 0.20 M acid, 0.10 M acetate.

$$\text{pH} = 4.74 + \log(0.10 \text{ M} / 0.20 \text{ M}) = 4.74 - 0.30 = \mathbf{4.44}$$

6. 0.20 M NH_3 , 0.10 M NH_4Cl .

$$\text{pH} = 9.25 + \log(0.20 \text{ M} / 0.10 \text{ M}) = \mathbf{9.55}$$

C. Adding acid

7. Add 0.020 mol HCl to 0.500 L of 0.20 M/0.20 M buffer.

Starting moles: acid = acetate = $0.20 \text{ mol/L} \times 0.500 \text{ L} = 0.10 \text{ mol}$.

After reaction: acetate = $0.10 \text{ mol} - 0.020 \text{ mol} = 0.08 \text{ mol}$; acetic acid = $0.10 \text{ mol} + 0.020 \text{ mol} = 0.12 \text{ mol}$.

$$\text{pH} = 4.74 + \log(0.08 \text{ mol} / 0.12 \text{ mol}) = 4.74 - 0.18 = \mathbf{4.56}$$

D. Challenge

8. Ratio for pH 5.00.

$5.00 = 4.74 + \log(\text{ratio})$, so $\log(\text{ratio}) = 0.26$ and $\text{ratio} = 10^{0.26} = \mathbf{1.8}$ (about 1.8 times as much acetate as acetic acid).